Tactix affine recovery and structured-root lifecycle repair

Engineering checkpoint for Andrew

October 4, 2026

Objective and outcome

This checkpoint evaluates the restored LymphoSAT affine constructor and the new partial-model guidance path on the focused regression subset. It also finishes three internal certified-UNSAT foundations and repairs the scheduler lifecycle identified by the terminal result.

The result is useful but not accepted. All six unique MDP instances previously solved directly by affine construction are recovered with independently checked original-CNF models. Three search-trajectory controls are not recovered: `cabp-V-nos6` is 4.6–5.1 times slower than the accepted reference, `Break_20_72` times out at 1,200 seconds, and the WV2 timetable row times out at 600 seconds. Partial-model guidance is never offered or consumed on any focused row, so this campaign establishes no guidance benefit.

The causal audit finds that affine had been restored at the wrong lifecycle boundary. The accepted source ran Lucky instance 8 as stage 0 of the structured-root scheduler. The first repair called it directly before `MaterializeColdState` and scheduler entry. Commit `f6d7453f6` restores the typed stage and removes the direct call. Focused normal and sanitizer tests pass; no benchmark in this report uses that new commit yet.

Exact campaign evidence

The control pins source `60446aa0d` and binary SHA-256 `262916d988bf98686005195ff8dac7d5c73fc46455c99d895c0b4c5b184b8820`. The guidance arm pins source `a49191bfa` and binary SHA-256 `7b959b074872d6245a0f8640535a4378bfcec6dfff86589f3aebdeb3f0a3bdc4`. The main solver cap is 1,200 seconds; WV2 keeps its 600-second cap. Independent checking is capped at 1,800 seconds. The historical comparison is not rerun.

Instance	Control	Guidance	Checked disposition
mdp-32-10	0.117	0.117	SAT; WV2 duplicate also checked
mdp-28-14	0.067	0.067	SAT
mdp-32-11	0.117	0.117	SAT
mdp-32-14	0.317	0.317	SAT
mdp-36-10	0.467	0.467	SAT
mdp-36-14	0.467	0.467	SAT
cabp-V-nos6	1,127.160	1,026.495	SAT, accepted reference 222.228
Break_20_72	timeout	timeout	accepted reference 134.586
WV2 timetable	timeout	timeout	accepted reference 482.234

Times are wall seconds. Every direct MDP task reports one affine call and one affine SAT completion. The three controls report one affine call and no completion. Both score files contain ten tasks, nine unique instances, eight passing tasks and two timeout failures; the repeated MDP row accounts for the tenth task.

All guidance telemetry is zero: offers, applications, Walk imports and imported phases, CDCL target installs and decisions, candidate replacement, and native, Rephase or Walk supersession. This is expected for the six direct solves, which terminate inside the constructor. The three controls produce no partial candidate. A useful matched experiment must first select rows where a candidate is actually published.

Both-policy fuzz

The initial jobs failed only because the reachability parser expected an older statistics spelling. Commit `666f8bad0` fixes the parser and adds a positive/zero reach regression. Replacement jobs use harness SHA-256 `3d98ab22c0e76aa500d65de9de16a9077a2833b1a2a10782ad19f0039f2e4a7b` and checker-set SHA-256 `895f1989468d5e01a81fa647f9cb9739448b81507a4b78515e4e46a7441c44f8`.

Policy/job	Corpus	Cases	Verdicts	Timeouts	Failures
Default / 1748417	51/51	10,000	9,718	282	0
Random / 1748418	51/51	10,000	8,464	1,536	0

The timeout cases receive no model, proof or differential oracle. Both jobs also report zero local-search-guidance applications. They are terminal zero-found-failure compatibility evidence, not complete 10,000-case oracle coverage or effective guidance fuzz acceptance.

Why the trajectories changed

Accepted `d170b4a5` and latest saved good `6fe14539` have the same `cabp` and Break decisions, conflicts, BVE work and structured-root output. Rejected `156afef6b` and recovered `60446aa0d` share the rejected prefix. On `cabp`, accepted execution applies no SR, eliminates 118,092 variables by BVE, and reaches 1,957,665 decisions and 86,308 conflicts. The recovered execution adds 1,350 unique-literal AMO binaries, eliminates 845,002 variables, and reaches 6,635,734 decisions and 873,391 conflicts. Break changes from

zero SR applications to 31 ULC binaries. Timetable changes from five SR applications and zero ULC binaries to 638 applications, 633 of them ULC binaries.

The earliest plausible change is `84e8d197a`, which removed affine from stage 0. Restoring only the affine call counter does not restore the accepted structured prefix. The exact-one counter also changed from two to one, but the removed second call ran after structured-root processing, so it cannot cause that earlier divergence. This is strong endpoint and source-order evidence, not a first-bad run at the intermediate commit.

Commit `f6d7453f6` now restores Lucky/8 as typed step 109 inside the structured-root inprocessing scope. The configured affine parameters are posed on that step, disabling affine removes it, and no second hidden call remains. A feedback-level regression pins stage order, no-op disposition, unchanged formula version, the expected strategy epoch transition, and later structural execution. The combined checkpoint suite passes 10/10 in both normal and ASan/UBSan builds, with leak detection disabled in the ptrace environment.

Certified-UNSAT foundations

Three internal verticals were hardened in parallel, without performance or target-recovery claims:

- GF(3) now has an independently replayed Gaussian-to-RUP certificate for unit rows and nonzero zero-row conflicts. Compact source references are bounded. There is no live publisher, nonunit consequence, auxiliary step or PB path.
- Direct PHP has exact live terminal DSR/DRAT publication and a Kernel-bound unregistered action/replay session. Proof-backed split preparation is rejected because the proof writer exposes no exact mutation epoch; one-shot apply/replay is atomic.
- Tseitin-cycle X7 has exact live replay and one DSR/no-proof root-unit publication from XOR groups authenticated by the caller. Autonomous group discovery and a public controller action remain absent.

The calculus and semantic-extension document now state these actual boundaries. Every decline remains bounded and mutation-free; none is promoted to a public learner-selectable capability yet.

Next actions and open gates

When benchmark execution is next authorized, run only `cabp`, Break and Timetable from `f6d7453f6`. Judge the structural prefix before wall time: `cabp` and Break must return to zero ULC binaries, while Timetable must return to five SR applications and zero ULC binaries. If that succeeds, proceed to the LymphoSAT-owned recovery subset and then B28+C5, Rank1–50, WV100 and WV2–89.

Full Debug, effective both-policy fuzz, the four current-source regression gates, all 44 independently checked targets, and a current N1–N4 audit remain open. The broader objective remains solving substantially more than LymphoSAT, Satsuma+Kissat, Kissat-sc2024 and AE-Kissat-MAB combined. The detailed evidence is in `benchmarks/reports/2026-10-04-affine-guidance-recovery.md`; its score JSON files bind every focused result.